HW5 11/8/24

1) 1. $f(x) = \frac{c}{x^2}$ $1 < x < \infty$

$c\int_1^\infty \frac{1}{x^2}dx = c\left[\frac{-1}{x}\right]_1^\infty = c(0+1) = 1$

$c = 1$

2. $E[X] = \int_1^\infty x f(x)dx = \int_1^\infty \frac{1}{x}dx = [\ln(x)]_1^\infty = \ln\infty - \ln(1) = \infty$

$\lim_{x\to\infty} \ln(x)$

Thus, not finite.

2) 1. $F(t) = \int_\delta^t \frac{1}{\theta} e^{-\frac{(x-\delta)}{\theta}} dx = \int_0^{\frac{t-\delta}{\theta}} e^{-u}du = \left[-e^{-u}\right]_0^{\frac{t-\delta}{\theta}} =$

Let $u = \frac{x-\delta}{\theta}$

$-e^{-\frac{t+\delta}{\theta}} + 1 = 1 - e^{-\frac{t+\delta}{\theta}}$

$\frac{du}{dx} = \frac{d}{dx}\left[\frac{x}{\theta}\right] - \frac{d}{dx}\left[\frac{\delta}{\theta}\right]$

$\frac{du}{dx} = \frac{1}{\theta}$

$dx = \theta du$

$\frac{1}{\theta} e^{-y/\theta}$

2. $y = x - \delta$

$E[x] = E[y] + \delta$

$E[x] = \theta + \delta$ mean

for $E[Y^2]$

$E[Y^2] - (E[Y])^2 = \theta^2$

$E[Y^2] - \theta^2 = \theta^2$

$E[Y^2] = 2\theta^2$

$E[X^2] - (E[X])^2 = E[(Y+\delta)^2] - (\theta+\delta)^2 =$

$E[Y^2 + 2\delta Y + \delta^2] - (\theta^2 + 2\theta\delta + \delta^2) =$

$E[Y^2] + 2\delta E[Y] + \delta^2 - \theta^2 - 2\theta\delta - \delta^2 =$

$2\theta^2 + 2\delta\theta - \theta^2 - 2\theta\delta = 2\theta^2 - \theta^2 = \theta^2$ variance

3) 1) 2 Ft in width / 3 flaws in 100 foot roll $\lambda = 3/110 = 0.03$

for 40 ft: $\lambda = 1.2$

$f(0) = \frac{\lambda^0 e^{-\lambda}}{0!} = 1.2^0 e^{-1.2} = e^{-1.2} = 0.301$

2) We are assuming that the flaws are of a uniform distribution. Also, that they're independent.

6) sum: $(-\lambda t + 1) + \left(\frac{-(\lambda t)^2}{2!} + \lambda t\right) + \left(\frac{-(\lambda t)^3}{3!} + \frac{(\lambda t)^2}{2!}\right) + \ldots + \left(\frac{-(\lambda t)^{w-1}}{(w-1)!} + \frac{(\lambda t)^{w-2}}{(w-2)!}\right)$

cont.

$\lambda e^{-\lambda t} - \lambda e^{-\lambda t}\left[1 - \frac{(\lambda t)^{w-1}}{(w-1)!}\right] = \lambda e^{-\lambda t} - \lambda e^{-\lambda t} + \frac{\lambda e^{-\lambda t}(\lambda t)^{w-1}}{(w-1)!} =$

$\frac{\lambda e^{-\lambda t}(\lambda t)^{w-1}}{(w-1)!} \square = \frac{\lambda(\lambda t)^{w-1} e^{-\lambda t}}{(w-1)!}$

4) Poisson. $\mu_x = \lambda = 16$ W = time in seconds b4 7th count made

sum of 7 random variables each indv one is exponential

$\theta = \frac{1}{16}$

1. $W \sim \text{Gamma}(\alpha = 7, \theta = \frac{1}{16})$
2. $1 - P(W > 0.5) = 1 - \sum_{k=0}^{6} \frac{8^k e^{-8}}{k!}$

hint: $F(w) = 1 - \sum_{k=0}^{\alpha-1} \frac{8^k e^{-8}}{k!}$

$= 1 - (e^{-8} + 8e^{-8} + \frac{64e^{-8}}{2} + \frac{512e^{-8}}{6} + \frac{4096e^{-8}}{24} + \frac{32768e^{-8}}{120} + \frac{262144e^{-8}}{720})$

$= 1 - (9e^{-8} + 32e^{-8} + 85.33e^{-8} + 170.66e^{-8} + 273.066e^{-8} + 364.088e^{-8})$

$= 1 - e^{-8}(937.144) = \boxed{0.6866} = P(W \leq 0.5)$

5) $M(t) = \frac{1}{(1-\theta t)^{\alpha}} = (1-\theta t)^{-\alpha}$

$M'(t) = -\alpha(1-\theta t)^{-\alpha-1}(-\theta) = \alpha\theta(1-\theta t)^{-\alpha-1}$

$M'(0) = \alpha\theta(1-\theta(0))^{-\alpha-1} = \alpha\theta$ □ ✓

$M''(t) = (-\alpha-1)(\alpha\theta)(1-\theta t)^{-\alpha-2}(-\theta) = (\alpha+1)(\alpha\theta^2)(1-\theta t)^{-\alpha-2}$

$M''(0) = (\alpha+1)(\alpha\theta^2)$

$\sigma^2_x = E[x^2] - (E[x])^2 = (\alpha+1)(\alpha\theta^2) - (\alpha\theta)^2 = \alpha^2\theta^2 + \alpha\theta^2 - \alpha^2\theta^2 =$

$\boxed{\alpha\theta^2}$ ✓

6) $1 - \sum_{k=0}^{w-1} \frac{(\lambda t)^k e^{-\lambda t}}{k!} = 1 - \left(e^{-\lambda t} + \sum_{k=1}^{w-1} \frac{(\lambda t)^k e^{-\lambda t}}{k!}\right)$

$= 1 - \left[e^{-\lambda t} + \sum_{k=1}^{w-1} \frac{1}{k!}(\lambda t)^k (e^{-\lambda t})\right]$ $\quad u = (\lambda t)^k \quad v = e^{-\lambda t}$

$F'(x) = 0 + \lambda e^{-\lambda t} - \frac{d}{dx}\sum_{k=1}^{w-1} \frac{1}{k!}(\lambda t)^k (e^{-\lambda t}) =$ $\quad u' = k(\lambda t)^{k-1}(\lambda) \quad v' = -e^{-\lambda t}(-\lambda) = \lambda e^{-\lambda t}$

$\lambda e^{-\lambda t} - \left(\sum_{k=1}^{w-1} \frac{1}{k!}\left[k(\lambda t)^{k-1}(\lambda)(e^{-\lambda t}) + (\lambda t)^k(-\lambda e^{-\lambda t})\right]\right) =$

$\lambda e^{-\lambda t} - \left[\sum_{k=1}^{w-1}\left(\frac{1}{k!}\right)(\lambda e^{-\lambda t})\left[k(\lambda t)^{k-1} - (\lambda t)^k\right]\right] = \lambda e^{-\lambda t} - \lambda e^{-\lambda t}\left\{\sum_{k=1}^{w-1} \frac{-(\lambda t)^k}{k!} + \frac{(\lambda t)^{k-1}}{(k-1)!}\right\}$ Cont. on next page